

LLM4DyG: Can Large Language Models Solve Spatial-Temporal Problems on Dynamic Graphs?

LLM4DyG (Zhang et al., KDD 2024) introduces the first benchmark to assess whether Large Language Models (LLMs) can **reason about dynamic graphs** — graphs that evolve over time — using **only natural language prompts**. Instead of relying on GNNs or fine-tuning, this work evaluates LLMs through zero- or few-shot question answering on serialized graph histories.

Problem Setup and Input Format

Each dynamic graph is represented as a list of timestamped edges:

[(1, 2, 0), (0, 1, 1), (2, 3, 2), (3, 4, 4)]

This means:

- Node 1 connects to node 2 at time 0
- Node 0 connects to node 1 at time 1
- Node 2 connects to node 3 at time 2
- Node 3 connects to node 4 at time 4

These serialized interactions are passed into the LLM in natural language form. The model is then asked questions such as:

- "When do node 1 and node 2 become connected?"
- "What are the neighbors of node 0 at time 1?"
- "Does a time-respecting path exist from node 1 to 4?"

Task Categories

Nine tasks are grouped into three categories:

Temporal Tasks

- *When Link*: Return the time when two nodes form a direct edge
- *When Connect*: Return the time when a path exists between two nodes
- *When Triadic Closure*: When do three nodes form a triangle?

Spatial Tasks

- *What Neighbor at Time*: List node's neighbors at a specific time
- *What Neighbor in Periods*: List neighbors after a given time
- *Check Triadic Closure*: Binary (Yes/No) question — did a triangle ever form?

Spatio-Temporal Tasks

- *Check Temporal Path*: Does a path exist with strictly increasing timestamps?
- *Find Temporal Path*: Find such a path
- *Sort Edges by Time*: Output edges ordered by their time

Models and Evaluation

Five LLMs are evaluated:

- GPT-3.5
- Vicuna (7B, 13B)
- LLaMA-2-13B
- CodeLlama-13B

They are tested on synthetic graphs generated by Erdos-Rényi (ER), stochastic block model (SBM), and Forest Fire, under varying sizes, densities, and time spans.

Key Findings

- **GPT-3.5 achieves the best overall accuracy**, solving up to 8 of 9 tasks with simple prompts
 - **CodeLlama** performs better than Vicuna or LLaMA-2 on reasoning-intensive tasks, likely due to its code reasoning ability
 - **Increasing graph size and density hurts performance** more than increasing time span
 - **Prompt engineering is crucial**: carefully designed prompts significantly improve accuracy
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DST² Prompting Strategy

The authors propose **DST² (Disentangled Spatial-Temporal Thoughts)** prompting:

First guide the model to think about time, then about graph structure (or vice versa)

Example:

"Step 1: Which edges occur before time 2? Step 2: Can a path be formed with those edges?"

This strategy improved performance substantially. For example, in the "When Link" task, GPT-3.5 accuracy jumped from **33.7% to 76.7%** with DST² prompting.

Why It Matters for the Survey

LLM4DyG is a landmark benchmark because it:

- **Tests LLMs on dynamic graph reasoning tasks** using only prompts
- **Avoids fine-tuning or architectural changes**
- Covers a wide range of reasoning types (temporal, structural, temporal+structural)
- Introduces a reusable evaluation protocol and prompt engineering method

It establishes a **baseline of what LLMs can and cannot do** with serialized graph inputs, and offers a useful diagnostic for LLM-GNN hybrid research. The work shows that **LLMs have**

potential as temporal-structural reasoners, but also clear limitations as graphs become complex.